

Learning Objective 2.2: I can simulate a dipole antenna using an EM simulation tool and interpret the results against analytical predictions.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Segmentation arithmetic.** You are about to model a straight center-fed dipole at $f = 2.45\text{GHz}$ in NEC. The wire is 0.8 mm in diameter, so the radius is $a = 0.4\text{ mm}$. Before touching the keyboard, size the model on paper.

- (a) Find the free-space wavelength and the physical length of a half-wave dipole at this frequency.

$$\lambda = \boxed{}$$

$$\lambda/2 = \boxed{}$$

- (b) You choose $N = 15$ segments. Find the segment length Δ and show that it satisfies both NEC bounds: $\Delta < \lambda/20$ and $\Delta > 8a$.

$$\Delta = \boxed{}$$

- (c) What is the largest *odd* segment count this wire will tolerate before it violates the $\Delta > 8a$ rule?

$$N_{\max} = \boxed{}$$

- (d) In one or two sentences, explain why the segment count must be odd.

2. **Predict, then account for the difference.** You model the 2.45 GHz dipole of Question 1 at *exactly* $\lambda/2 = 61.2$ mm, with 15 segments and a 1 V source on the center segment. NEC returns $Z_{\text{in}} = 86.9 + j46.2\Omega$ and a peak gain of 2.17 dBi.

(a) State the four predictions Lesson 7 makes for a half-wave dipole: input impedance at exactly $\lambda/2$, resonant length, input resistance at resonance, and directivity in dBi.

(b) Compute the percent difference between the simulated and predicted values of R_{in} and of X_{in} .

R_{in} error =

X_{in} error =

(c) In two or three sentences, account for the resistance difference. Name the mechanism; “simulation error” is not an explanation.

(d) Trimming brings this wire to resonance at 0.473λ . Find the resonant length in millimetres and the percent shortening from $\lambda/2$.

L_{res} =

trim =

3. **Reading a convergence study.** A classmate models a dipole of length exactly 0.5λ with wire radius $a = 0.001 \lambda$ and re-runs it at five segment counts:

Segments N	$R_{\text{in}} (\Omega)$	$X_{\text{in}} (\Omega)$
5	79.9	+35.9
11	81.9	+43.9
21	83.4	+45.6
41	84.5	+46.5
81	85.4	+47.2

- (a) Compute the percent change in R_{in} from each row to the next.

last step =

- (b) Find the largest segment count the $\Delta > 8a$ rule permits for this wire. Which rows of the table violate it?

$N_{\text{max}} =$

- (c) Which segment count would you defend for this model, and why? Your answer must reference both the convergence trend and the kernel limit.

- (d) Explain in one or two sentences why gain converges at a lower segment count than input impedance does.

4. **The average-gain audit.** After a full-sphere pattern run of a lossless dipole in free space, NEC reports an average power gain of 1.15.

(a) What does an average power gain of exactly 1.000 tell you about a lossless free-space model, and why is that a test rather than a result?

(b) Express the reported 1.15 in decibels, and state whether you would accept the run.

avg gain =

(c) Give two plausible causes of an average gain this far from 1.000, and the specific check that would confirm each.

5. **Why the current comes first.** A classmate claims that NEC “computes the radiation pattern directly from the geometry.”

(a) In two or three sentences, explain why a method-of-moments code must solve for the current distribution before it can produce a pattern. Refer to the radiation integral.

- (b) The solver drives the center segment with 1 V and reports a feed current of $8.90 - j4.80$ mA. Find Z_{in} , then the magnitude of the reflection coefficient and the VSWR referred to 50Ω .

$$Z_{in} = \boxed{}$$

$$|\Gamma| = \boxed{}$$

$$\text{VSWR} = \boxed{}$$

- (c) You need this antenna matched to 50Ω . Would you trim the wire to resonance or add a matching network? Give the trade-off in one or two sentences.

Documentation: _____